

Production lines — Basic design

Document control

Field	Value
Document ID / version	005_BD / 1
Category / system / subsystem	UI / ProductionManagementAI / Master data
Work item / based on requirements	WI-009 / 005_REQ version 1
Author / created / updated	Agent / 2026-10-01 / 2026-10-01
Review state	Submitted for review; not implemented

Version	Date	Author	Change
1	2026-10-01	Agent	Initial design from confirmed DEC-001–DEC-009

System overview

SCR-005 maintains production lines, working hours/day and supported products with minutes per one product unit. Admin and Operator maintain these masters. Orders may omit a line in Draft but need an eligible line to begin production. This feature describes capacity; it does not schedule, allocate or detect overload.

Requirement	Design coverage
REQ-064	List, search, paging and visible states
REQ-065	Identity form, uniqueness and concurrency
REQ-066	Hours and product timing editor; unit reconfirmation
REQ-067	Retirement confirmations and historical assignment
REQ-068	Order integration and assignment lock
REQ-069	Role checks, Japanese UI, navigation and accessibility

[Requirements](#) and [decisions](#) are the business sources.

Overall configuration and architecture

Reuse React, the Japanese catalog, centralized Lucide icons and same-origin ASP.NET Core controllers, Application ports/use cases and EF/PostgreSQL storage. Existing [layering ADR](#) and [authentication ADR](#) remain unchanged; [SA assessment](#) requires no new ADR. Proposed master API base: `/api/production-lines`. Exact endpoint/schema/transaction contracts belong to the later DB/API/FN designs.

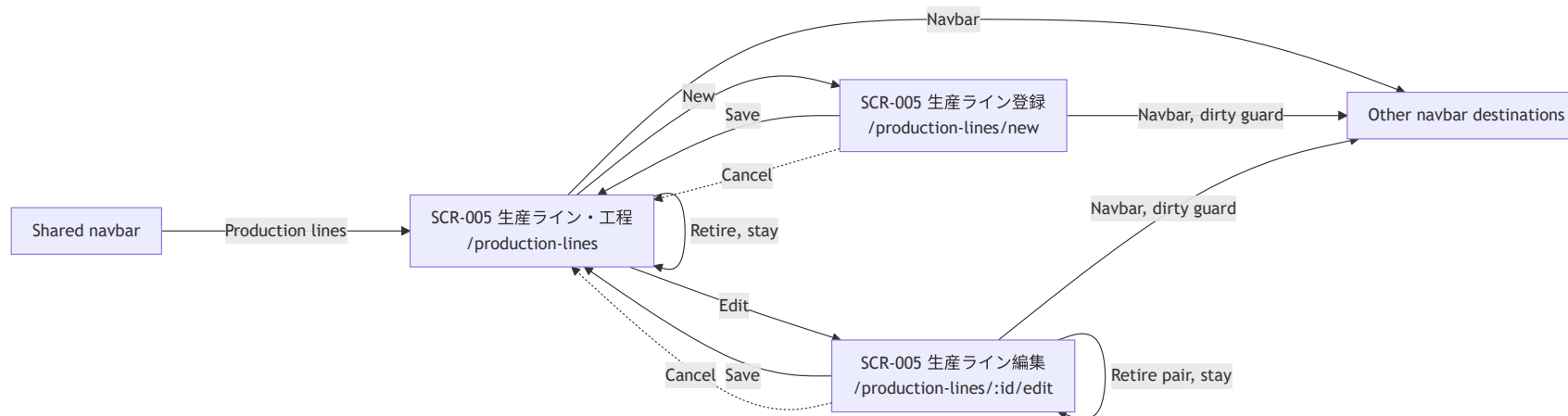
Function list and business flow

ID	Function	Requirements
FN-032	Search and page lines	REQ-064
FN-033	Create/edit line identity and working hours	REQ-065, REQ-066
FN-034	Maintain product support, time coefficient and unit confirmation	REQ-066, REQ-067
FN-035	Retire a line with history preserved	REQ-067
FN-036	Select/validate/lock a line on an order	REQ-068, REQ-069

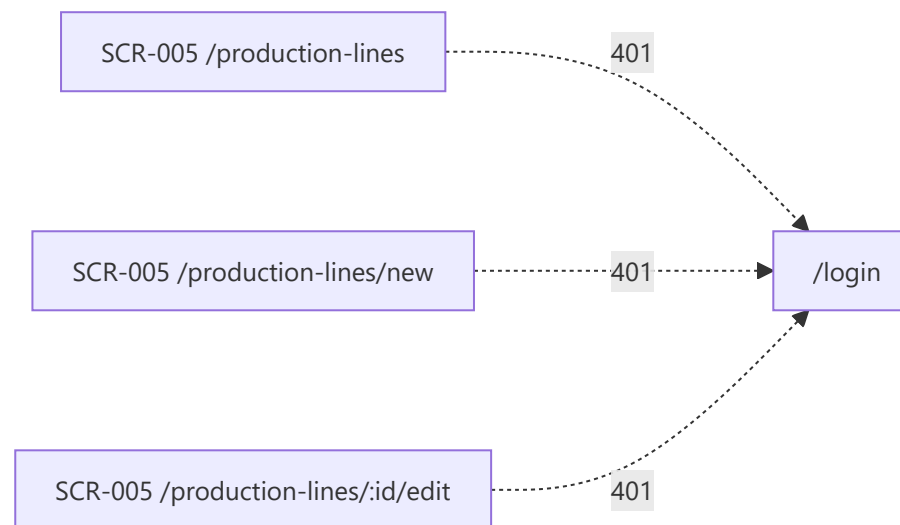
1. Admin/Operator opens 「生産ライン・工程」 (Production lines and processes), searches code/name, filters state and opens create or edit.
2. Enter immutable code at creation, mutable name and working hours/day; add supported active products with minutes per unit. A line may initially have no supported products; it is then unavailable to any order selection.
3. Save line and configured pair changes together. Validation leaves the dirty form intact; success returns to the list with a visible/announced notice.
4. Retire a line or a saved product association only after confirmation. Rows and historical references remain; unsaved pair rows can be removed locally.
5. On an order, choose a line eligible for its product; begin production only after server-side eligibility checks. No scheduling or capacity reservation.

Screen list and transition

Screen/mode	Entry	Exit / next state
SCR-005 list, <code>/production-lines</code>	Shared navbar; form Save/Cancel	New, Edit, Retire stays on list, other navbar destinations, <code>/login</code> on 401
SCR-005 create, <code>/production-lines/new</code>	List New	Save/Cancel to list; other navbar destinations after dirty guard; <code>/login</code> on 401
SCR-005 edit, <code>/production-lines/:id/edit</code>	List Edit	Save/Cancel to list; pair retirement stays on edit; other navbar destinations after dirty guard; <code>/login</code> on 401



Session expiry exits (same modes, separated for legible arrows):



Dialog is a mode overlay, not a route. Cancel on clean forms exits directly; leave/reload on dirty forms requires discard confirmation. A 403 stays on the current route in a forbidden state without another write.

Screen design detail — SCR-005

0-1. Basic information

Item	Design
Routes / API base	Three routes above; proposed <code>/api/production-lines</code>
Encoding / responsive	UTF-8; PC table and SP cards; shared breakpoint convention
Authentication / write roles	Cookie session; Admin or Operator enforced at API and UI
Error fallback	Local retry/404/403 state; <code>/login</code> on 401
Query / route parameters	List <code>q</code> , <code>state</code> , <code>page</code> ; edit <code>id</code> UUID; apply/reset starts page 1

0-2. Page metadata

Title: current mode heading followed by ProductionManagementAI. Base, link, meta, style and script overrides: not applicable; shared document/bundles apply. No public indexing, social-preview content or external script required.

0-3. URL behavior

Applied filters use the URL and browser navigation restores them. Unsaved field values are local state, not URL parameters. Invalid page/state/UUID produces bounded validation or not-found state; exact query limits defined in API DD.

1. Layout and mockup

① ProductionManagementAI

ダッシュボード 製造指示一覧 製品マスタ 生産ライン・工程

② 生産ライン・工程

新規ライン

③ 検索

ラインコード・ライン名 状態 検索 クリア

④ 一覧

コード ライン名 稼働時間／日 状態

L-001 第1ライン 8時間 使用中

編集 使用停止

⑤ ページ・状態

全件数 前へ 1 次へ

⑥ 生産ライン編集

ラインコード : L-001 (変更不可)

ライン名 : 第1ライン 稼働時間／日 : 8

保存 キャンセル

⑦ 生産可能な製品

製品 単位 製造時間／単位 (分)

P-1001 個 2.5 使用中

製品を追加 使用停止

⑧ 単位の確認が必要です

前の単位 : 個 現在の単位 : kg

現在の単位で確認

⑨ 確認ダイアログ

このラインを使用停止にしますか？

キャンセル 使用停止

① PMAI
メニュー

② 生産ライン・工程
新規ライン

③ 検索
ラインコード・ライン名
状態 検索 クリア

④ 一覧カード
L-001 第1ライン
稼働時間／日：8 時間 使用中
編集 使用停止

⑤ ページ・状態
全件数 前へ 1 次へ

⑥ 生産ライン編集
ラインコード：L-001（変更不可）
ライン名：第1ライン
稼働時間／日：8 保存 キャンセル

⑦ 生産可能な製品
P-1001 単位：個
製造時間／単位：2.5 分／個
製品を追加 使用停止

⑧ 単位の確認が必要です
前の単位：個 現在：kg
現在の単位で確認

⑨ 確認ダイアログ
使用停止しますか？
キャンセル 使用停止

These are design wireframes, not screenshots of a running application. Regions 6–9 show form/overlay states alongside the list for reference, not one combined product screen. English/Japanese HTML mockups belong to the later SPD package.

No.	Region	Content and behavior
1	Header	Shared navigation, new line entry, current marker; mobile menu
2	Heading/action	Mode heading; list New; form heading and dirty-state notices
3	Search/filter	Code/name search, active-state filter, Search/Clear
4	Results	Code, name, hours/day, state, Edit and active-line Retire; table/cards
5	List status/paging	Total, page, previous/next; loading/empty/error/success notices
6	Identity form	Code (immutable on edit), name, hours/day, validation, Save/Cancel
7	Product timing	Supported product/unit, minutes per unit, Add and retirement; unsaved remove
8	Unit reconfirmation	Stale-unit warning, previous/current unit, explicit confirmation action
9	Confirmation	Centered native dialog for line/pair retirement or discarding draft

2. Content blocks and visible states

State	Visible response
Loading / empty	Loading status; distinguish no lines from no search matches; New remains available
Ready / retired	Rows/cards; active/retired text, not color alone; retired lines/pairs remain readable
Fetch / 404 / 403	Local Retry for transient failure; not-found or forbidden view; no invented/stale data presented as current
Validation	Summary plus field errors; focus first invalid field; retain all entered rows/values
Conflict	Explain changed record; explicit reload guarded against dirty-data loss; never overwrite silently
Unit changed	Timing row shows configured/current units and cannot be newly used until explicit confirmation
Saving / success	Disable repeated writes while pending; announce success and return to list
Uncertain write	Preserve entered data; explain outcome unknown; offer list/reload verification, no automatic replay

3. Screen items

The Japanese labels below are proposed catalog additions. Existing order labels remain unchanged. Technical text length limits follow master conventions and are reviewable proposals; exact API/DB mapping is finalized later.

Item	Japanese UI (English gloss)	Rule
Line code	「ラインコード」 (Line code)	Required, trim, proposed max 50 Unicode code points; immutable; case-insensitive unique even retired
Line name	「ライン名」 (Line name)	Required, trim, proposed max 200 code points
Working hours	「稼働時間／日」 (Working hours/day)	0 < hours <=24; <=3 fractional digits; unit 「時間」 (hours)
Supported products	「生産可能な製品」 (Supported products)	Choose active products; one saved association per line/product; saved retired rows kept for history
Product timing	「製造時間／単位」 (Production time/unit)	Positive minutes per one actual product unit, <=3 fractional digits; no implicit batch/unit conversion
Pair state	「使用中」 / 「使用停止」 (Active / Retired)	Retire confirmed; no hard delete or reactivation in this scope
Unit warning	「単位の確認が必要です」 (Unit confirmation required)	Show previous/current unit; change/confirm coefficient for current unit explicitly before Save
Confirm unit	「現在の単位で確認」 (Confirm for current unit)	Enabled after a valid current-unit coefficient; never auto-confirm on load/save
Actions	「新規ライン」 / 「編集」 / 「保存」 / 「キャンセル」 (New line / Edit / Save / Cancel)	Save line and pair edits together; Cancel does not write
Pair actions	「製品を追加」 / 「使用停止」 (Add product / Retire)	Unsaved association can be locally removed; persisted retirement is confirmed

4. Item value mapping

Display hours and minutes with exact decimal formatting and visible units; never sum unlike product-unit capacities. `IsActive` true/false maps to active/ retired Japanese text. A unit mismatch adds the confirmation-required state, independent of retirement. Missing historical order line displays 「未設定」 (Unassigned), not an invented default. Proposed names/labels are not implemented.

5. Validation and history rules

- API repeats all identity, precision, role and eligibility checks; UI validation is assistance, not an authorization boundary. Code remains immutable via API.
- Configured timing is tied to its confirmed product unit. Product-unit edits permitted by WI-006 remain permitted; old-unit timing is not reinterpreted. Explicit current-unit reconfirmation is required for new selection/production start.
- New/changed assignment requires active product, active line and active pair with a matching confirmed unit. Draft -> InProgress applies this check even if the Draft retained a historically retired assignment. Explain how to choose a valid line.

- Unchanged historical assignments may be retained during otherwise allowed edits. Outside Draft, reject assignment changes; legacy null stays null. New Drafts may omit a line; no default/backfill or change to other existing field locks.
- Current timing edits do not recalculate history. No historical duration snapshot or derived schedule promised. Retiring a saved pair/line retains its identity.
- Pending local edits are validated together; when a retirement affects dirty data, require explicit save/discard resolution instead of silently losing edits.

6. Item events

Event	Result
Navbar / Search / Clear / Page	Enter line list; apply/reset URL filters; fetch bounded page
New / Edit	Enter respective form; focus heading after mount
Add product	Append unsaved timing row; selectable products exclude duplicates/inactive products
Save	Validate; send one logical line/pair edit; retain on failure, list on confirmed success
Retire line/pair	Open centered confirmation naming target; Cancel/Escape no write; confirmed line stays list, pair stays edit
Unit confirmation	User explicitly confirms timing for visible current unit; save guarded against another unit-change race
Cancel / dirty navigation / Reload	Discard confirmation when dirty; Keep editing retains draft; Escape safe; return focus

7. External identity linkage

Not applicable: no new identity provider, external ID synchronization or SSO.

Order integration boundaries

SCR-001 adds 「生産ライン」 (Production line) selection in Draft and read-only display otherwise. SCR-002 shows assigned line or 「未設定」 (Unassigned); line filters, dashboard by line and schedules are excluded. Product change in Draft clears/revalidates an incompatible line. Required validation is on beginning production, not Draft creation. Legacy orders already outside Draft with no line remain usable under existing field rules. A new WI-009-owned order-impact DD will specify states/requests without editing prior approved screen designs.

Accessibility, security and nonfunctional behavior

Target WCAG 2.2 AA: semantic table/cards, associated labels/errors, live statuses, keyboard-visible focus, 44-pixel touch action targets where practical and usable 200% zoom. Native confirmations fit viewport with 16-pixel margins, centered both axes including scrolled screens; initial safe action focused, background inert, Escape/Cancel safe and focus restored. Dirty dialogs follow same rules. Use a dedicated centralized `Factory` glyph (proposal), distinct from Package and ClipboardList, decorative aria-hidden with visible text/current marker.

Never log credentials, names/raw search or timing values in metric labels. Exact bounded operation/outcome spans belong to API/FN DD. Retire/selection and unit reconfirmation races require transaction-safe server checks; additive schema/FK preservation and recovery limits belong to DB design. No deployment gate claimed.

Traceability and next review

Check group	Acceptance / later verification
List/forms	REQ-064/065; filters, uniqueness, conflicts, uncertain writes
Timing/units	REQ-066; hours/time precision, unit reconfirmation and racing edits
Retirement/orders	REQ-067/068; no cancellation writes, historical exceptions, status lock and eligible start
Access/mobile	REQ-069; role API/UI checks, axe/keyboard/scroll/zoom/dialog geometry/navigation

This package is a design proposal, not working application evidence. Submit 005_BD with both PDFs and rendered wireframes; wait for explicit user instruction before writing 005_DB or any subsequent design Markdown. Application code waits for the separately approved implementation plan.